

SOP-992: Edge Analytics and Predictive Maintenance Model Deployment

Document Owner: Engineering Team

Last Updated: October 2025

This document outlines the standard operating procedure for deploying updated machine learning models to our factory floor gateways. Historically, this was done manually via USB drives, but we have recently integrated new telemetry systems to handle this over the air.

The process begins when a Data Scientist finalizes a new predictive maintenance model. They compile the model using Python and evaluate its accuracy against the baseline logs. Before doing anything else, the model's memory footprint must be evaluated. If the compiled model exceeds 50MB, the Data Scientist is required to run the quantization script to reduce its size.

Once the size is within acceptable limits (either originally or post-quantization), the model artifact is uploaded to the GitLab Package Registry.

At this point, the deployment is paused. Nothing can proceed until the Plant Manager logs into Jira and officially approves the weekly maintenance window ticket. Sometimes, the manager rejects it if production quotas are behind; in that case, the process stops entirely and the ticket is rescheduled for the following week.

Assuming Jira approval is granted, the DevOps Engineer takes over during the specified window. They trigger a deployment pipeline using AWS IoT Core. The cloud system attempts to push the new model directly to the local edge gateways. If the edge gateway is offline or rejects the payload due to a network timeout, the DevOps Engineer must manually SSH into the backup PLC network to force the update via terminal.

Once the new model is successfully running on the hardware, it automatically begins streaming anomaly data to our central Java backend. The backend system will then update the Grafana dashboards in real-time. Finally, a custom Slack bot posts a deployment summary to the #factory-ops channel, notifying all stakeholders that the process is complete.